

OCL Constraints for Study Group Platform

The following OCL constraints are designed specifically for the Study Group Platform project. All constraints correspond to entities, attributes, and services implemented in the project. The formatting follows the same academic style used in the teaching assistant slides.

1. Unique User Email

```
context User

inv UniqueEmail:
User.allInstances()->isUnique(u | u.email)
```

2. Group Maximum Members Limit

```
context Group

inv MaxMembersLimit:
self.members->size() <= self.maxMembers
```

3. Post Author Must Be Group Member

```
context Post

inv AuthorMustBeMember:
self.group.members
->exists(m | m.userId = self.authorId)
```

4. Unique Join Requests

```
context JoinRequest

inv UniqueJoinRequest:
JoinRequest.allInstances()
->isUnique(j | Tuple{
group = j.groupId,
student = j.studentId
})
```

5. Group Creator Must Be a Member

```
context Group

inv CreatorIsMember:
```

```
self.members  
->exists(m | m.userId = self.creatorId)
```

6. Valid Meeting Type

```
context Group  
  
inv ValidMeetingType:  
self.meetingType = 'ONLINE' or  
self.meetingType = 'OFFLINE'
```

7. Join Group Precondition

```
context GroupService::joinGroup(  
  groupId : UUID,  
  studentId : UUID  
)  
  
pre GroupNotFull:  
self.group.members->size() <  
self.group.maxMembers
```

8. Accept Request Postcondition

```
context JoinRequestService::acceptRequest()  
  
post StatusChanged:  
self.status@pre = 'PENDING'  
implies self.status = 'ACCEPTED'
```